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Chapter 2

NFA Revision

Automata

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String Acceptance

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① String Acceptance

② Regex $\rightarrow$ NFA

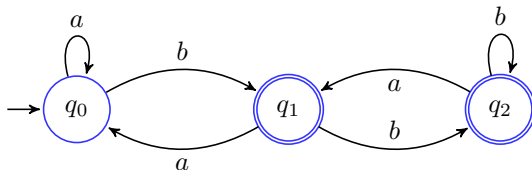
③ Language description $\rightarrow$ NFA

④ NFA $\rightarrow$ Language Description

Exercise

Let $\Sigma = \{a, b\}$

- 1 Does the given NFA accept the string "ab"?
- 2 Does the NFA accept "aba"?
- 3 Does the NFA accept "bbb"?
- 4 Does the NFA accept the empty string ε ?
- 5 Does the NFA accept "aab"?
- 6 Does the NFA accept "baba"?
- 7 Does the NFA accept "aaab"?
- 8 Does the NFA accept "abb"?
- 9 Does the NFA accept "ba"?
- 10 Does the NFA accept "abab"?





Let $\Sigma = \{a, b, c\}$

- Construct an NFA for the regex $E_1 = (a + b)^*ab$,
- Construct an NFA for $E_2 = a(b + c)^*$,
- Construct an NFA for $E_3 = (ab + ba)^*$,
- Construct an NFA for $E_4 = (a + bc)^*$,
- Construct an NFA for $E_5 = a^*b^*$,
- Construct an NFA for $E_6 = (a + b)^*aa$,
- Construct an NFA for $E_7 = ab(a + b)^*$,
- Construct an NFA for $E_8 = (ab)^* + (ba)^*$,
- Construct an NFA for $E_9 = (a + b)^*(ab)(a + b)^*$,
- Construct an NFA for $E_{10} = (a + b)^*b(a + b)^*b$.



Let $\Sigma = \{a, b, c\}$

- ① Construct an NFA for strings containing at least one 'a'
- ② Construct an NFA for strings ending with 'b'
- ③ Construct an NFA for strings starting with 'a'
- ④ Construct an NFA for strings containing substring "ab"
- ⑤ Construct an NFA for strings not containing "bb"
- ⑥ Construct an NFA for strings with exactly two 'a' characters
- ⑦ Construct an NFA for strings with even number of 'b'
- ⑧ Construct an NFA for strings with length ≥ 2
- ⑨ Construct an NFA for strings containing "aba"
- ⑩ Construct an NFA for strings without 'a'.

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Exercise

Let $\Sigma = \{a, b\}$

Give 3 valid strings & 5 invalid strings in language L^2 , with L represented by the following NFA, and then give the language description.

